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Display apparatus and panel with fan-out wiring reducing border width

Abstract

A display apparatus and a display panel are disclosed. The display panel includes a display area, a driving area, and a fan-out area provided with a plurality of fan-out wirings and at least including a first section wiring connected with data signal terminals, a second section wiring connected with the first section wiring, and a third section wiring connected with the second section wiring. The fan-out wirings are close to a central axis of the driving area, so that a difference between lengths of the first section wirings on both sides and a length of the first section wiring close to the central axis is reduced, thus effectively shortening the length of the first section wiring close to the central axis when the first section wiring on both sides maintains a predetermined length, therefore, the fan-out area is compressed and a width of the display panel border is shortened.

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Background/Summary

RELATED APPLICATION

(1) This application claims the benefit of priority of Chinese Patent Application No. 202211262381.1 filed on Oct. 14, 2022, the contents of which are incorporated by reference as if fully set forth herein in their entirety.

FIELD AND BACKGROUND OF THE INVENTION

(2) The present application relates to a technical field of display, and specifically to a display apparatus and a display panel thereof.

(3) In related technologies, output wirings in an output area of a display panel generally adopts a way shown in FIG. 1. One end of a straight fan-out wiring 210' is directly connected with a data signal line 110', and another end is directly connected with a data signal terminal 310', so as to realize connection between the data signal terminal 310' and the data signal line 110'. However,

with technical development of a full-screen display screen, screen proportion of the display screen is constantly increasing, and a lower border of the display screen is becoming less and less. This wiring mode has become one of factors that restrict narrowing of the lower border of the display screen.

SUMMARY OF THE INVENTION

(4) The embodiment of the application provides a display apparatus and a display panel thereof, which can narrow a lower border without affecting resistance of fan-out wirings.

(5) A display panel is provided in an embodiment of the present application, the display panel at least comprises a display area, a fan-out area, and a driving area; the fan-out area is arranged on one side of the display area, the driving area is arranged on one side of the fan-out area away from the display area, and the fan-out area is connected with the driving area and the display area; the display panel further comprises: a plurality of data signal lines located in the display area; a plurality of data signal terminals located in the driving area; and a plurality of fan-out wirings located in the fan-out area; the fan-out wirings are connected with the data signal lines and the data signal terminals in one-to-one correspondence; wherein, the fan-out wirings at least comprise a first section wiring, a second section wiring, and a third section wiring; the first section wiring is connected with the data signal terminals, the third section wiring is connected with the data signal lines, and the second section wiring is connected with the first section wiring and the third section wiring; the third section wiring extends from the display area to the driving area along a direction close to a central axis of the display panel, an extending direction of the first section wiring and an extending direction of the second section wiring are different, and the extending direction of the second section wiring and an extending direction of the third section wiring are different.

(6) In some embodiments of the present application, an orthogonal projection length of each third section wiring on a reference line coplanar with the third section wiring is less than a distance between adjacent two data signal lines; wherein an extending direction of the reference line is perpendicular to an extending direction of the central axis of the display panel.

(7) In some embodiments of the present application, each fan-out wiring further comprises a fourth section wiring, and the fourth section wiring is connected with the second section wiring and the third section wiring; the fourth section wiring of each fan-out wiring is arranged in parallel along a first direction, and the first direction is directed from the display area to the driving area.

(8) In some embodiments of the present application, a first endpoint of the fourth section wiring of each fan-out wiring is located on a first straight line extending along a second direction, a second endpoint of the fourth section wiring of each fan-out wiring is located on a second straight line extending along the second direction, and the second direction intersects the first direction.

(9) In some embodiments of the present application, the plurality of fan-out wirings comprise a plurality of fan-out wiring groups symmetrical about the central axis of the display panel; in at least one fan-out wiring group, the second section wiring of each fan-out wiring is arranged in parallel, and the central axis coincides with a central line of the fan-out area and the driving area.

(10) In some embodiments of the present application, in at least one fan-out wiring group, in the direction close to the central axis, lengths of the first section wirings of at least two fan-out wirings increase progressively.

(11) In some embodiments of the present application, in at least one fan-out wiring group, the first section wirings of the fan-out wirings are arranged at equal intervals, the second section wirings of the fan-out wirings are arranged at equal intervals, and the fourth section wirings of the fan-out wirings are arranged at equal intervals.

(12) In some embodiments of the present application, in at least one fan-out wiring group, a first distance between the first section wirings of adjacent two fan-out wirings is less than a second distance between the fourth section wirings of the adjacent two fan-out wirings, and the second distance is less than a third distance between wiring portions of the adjacent two fan-out wirings.

(13) In some embodiments of the present application, a fourth distance of the second section

wirings of the adjacent two fan-out wirings is less than the first distance of the first section wirings of the adjacent two fan-out wirings.

(14) Accordingly, a display apparatus is further provided in the embodiment of the present application, the display apparatus comprises: a shell; a display panel mounted on the shell and at least comprising a display area, a fan-out area, and a driving area; the fan-out area is arranged on one side of the display area, the driving area is arranged on one side of the fan-out area away from the display area, and the fan-out area is connected with the driving area and the display area; the display panel further comprises: a plurality of data signal lines located in the display area; a plurality of data signal terminals located in the driving area; and a plurality of fan-out wirings located in the fan-out area; the fan-out wirings are connected with the data signal lines and the data signal terminals in one-to-one correspondence; wherein, the fan-out wirings at least comprise a first section wiring, a second section wiring, and a third section wiring; the first section wiring is connected with the data signal terminals, the third section wiring is connected with the data signal lines, and the second section wiring is connected with the first section wiring and the third section wiring; the third section wiring extends from the display area to the driving area along a direction close to a central axis of the display panel, an extending direction of the first section wiring and an extending direction of the second section wiring are different, and the extending direction of the second section wiring and an extending direction of the third section wiring are different.

(15) In some embodiments of the present application, an orthogonal projection length of each third section wiring on a reference line coplanar with the third section wiring is less than a distance between adjacent two data signal lines; wherein an extending direction of the reference line is perpendicular to an extending direction of the central axis of the display panel.

(16) In some embodiments of the present application, each fan-out wiring further comprises a fourth section wiring, and the fourth section wiring is connected with the second section wiring and the third section wiring; the fourth section wiring of each fan-out wiring is arranged in parallel along a first direction, and the first direction is directed from the display area to the driving area.

(17) In some embodiments of the present application, a first endpoint of the fourth section wiring of each fan-out wiring is located on a first straight line extending along a second direction, a second endpoint of the fourth section wiring of each fan-out wiring is located on a second straight line extending along the second direction, and the second direction intersects the first direction.

(18) In some embodiments of the present application, the plurality of fan-out wirings comprise a plurality of fan-out wiring groups symmetrical about the central axis of the display panel; in at least one fan-out wiring group, the second section wiring of each fan-out wiring is arranged in parallel, and the central axis coincides with a central line of the fan-out area and the driving area.

(19) In some embodiments of the present application, in at least one fan-out wiring group, in the direction close to the central axis, lengths of the first section wirings of at least two fan-out wirings increase progressively.

(20) In some embodiments of the present application, in at least one fan-out wiring group, the first section wirings of the fan-out wirings are arranged at equal intervals, the second section wirings of the fan-out wirings are arranged at equal intervals, and the fourth section wirings of the fan-out wirings are arranged at equal intervals.

(21) In some embodiments of the present application, in at least one fan-out wiring group, a first distance between the first section wirings of adjacent two fan-out wirings is less than a second distance between the fourth section wirings of the adjacent two fan-out wirings, and the second distance is less than a third distance between wiring portions of the adjacent two fan-out wirings.

(22) In some embodiments of the present application, a fourth distance of the second section wirings of the adjacent two fan-out wirings is less than the first distance of the first section wirings of the adjacent two fan-out wirings.

(23) In the embodiments of the present application, fan-out wirings with fewer bends in prior art are modified into the fan-out wirings with lots of bends, so that the fan-out wirings are close to a

central axis of the driving area, so that a difference between lengths of the first section wirings on both sides and a length of the first section wiring close to one side of the central axis is reduced, thus effectively shortening the length of the first section wiring close to the one side of the central axis when the first section wiring on both sides maintains a predetermined length, so that the fan-out area is compressed, and the driving area can be set closer to the display area, which effectively shortens a width of the display panel border. To some extent, it solves a technical problem of the wiring mode in the prior art that affects narrowing of the lower border of a display, and realizes the narrowing of the lower border without affecting a wiring width and wiring resistance. After the fan-out area is compressed, space on both sides of the fan-out wirings increases, which can increase wiring space of other wirings, and appropriately increase the wiring width of other wirings to reduce the resistance.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) To illustrate technical solutions in an embodiment of the present application more clearly, accompanying drawings used in a description of the embodiments are briefly introduced as follows. Obviously, the accompanying drawings in the following description are only some embodiments of the present application. For those skilled in the art, other drawings can also be obtained from these drawings without creative effort.

(2) FIG. 1 is a wiring arrangement schematic diagram of a display panel in the prior art in the present application.

(3) FIG. 2 is a structural schematic diagram of a display apparatus provided in an embodiment of the present application.

(4) FIG. 3 is a wiring arrangement schematic diagram of a display panel provided in one embodiment of the present application.

(5) FIG. 4 is a wiring arrangement schematic diagram of the display panel provided in another embodiment of the present application.

(6) FIG. 5 is a wiring arrangement schematic diagram of the display panel provided in another embodiment of the present application.

(7) FIG. 6 is a wiring arrangement schematic diagram of the display panel provided in another embodiment of the present application.

DESCRIPTION OF DRAWINGS LABELS

(8) **10** shell; **20** display panel; **100,100'** display area; **200,200'** fan-out area; **300,300'** driving area; **400** electrostatic discharge protection area; **110,110'** data signal line; **111** main portion; **112** wiring portion; **120** common signal line; **130** control signal line; **210,210'** fan-out wiring; **220** common wiring; **230** signal control wiring; **240** signal conversion circuit; **250** conversion wiring; **201** first fan-out wiring group; **202** second fan-out wiring group; **203** common wiring group; **204** signal control wiring group; **211** first section wiring; **212** second section wiring; **213** third section wiring; **214** fourth section wiring; α first fan-out angle; β second fan-out angle; **310,310'** data signal terminal; **320** common signal terminal; **351** central axis; **330** control signal terminal; **350,350'** driving circuit.

DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

(9) Technical solutions in embodiments of the present application will be clearly and completely described below with reference to accompanying drawings in the embodiments of the present application. Obviously, described embodiments are only some, but not all embodiments of the present application. Based on the embodiments of the present application, all other embodiments obtained by those skilled in the art without creative efforts shall fall within protection scope of the present application. In addition, it should be understood that the specific embodiments described

here are only used to explain and interpret the present application and are not used to limit the present application. In the present application, location words used, such as “up” and “down”, generally refer to up and down of a device in actual use or working state, in particular the drawing direction in the drawings, without any description to the contrary; and the words “inside” and “outside” refer to an outline of the device.

(10) A display apparatus and a display panel **20** thereof are provided in the embodiments of the present application. The details are as follows. It should be noted that, a description order of the following embodiments is not used to define a preferred order of the embodiments.

(11) A display apparatus is provided in the present application, and the display apparatus can be mounted on products with lots of display functions. For example, electronic products can be intelligent terminals, laptops, photographic equipment, wearable devices, electronic scales, vehicle mounted displays, televisions, etc.

(12) Please refer to FIG. 1, FIG. 1 is a wiring arrangement schematic diagram of a display panel in the prior art in the present application. In the prior art, the display panel includes a display area **100'**, a fan-out area **200'**, and a driving area **300'**. The fan-out area **200'** is arranged on one side of the display area **100'**, the driving area **300'** is arranged on one side of the fan-out area **200'** away from the display area **100'**, and the fan-out area **200'** is connected with the driving area **300'** and the display area **100'**.

(13) The display panel further includes a plurality of data signal lines **110'**, a plurality of data signal terminals **310'**, and a plurality of fan-out wirings **210'**. The data signal lines **110'** and the fan-out wirings **210'** are connected with the data signal terminals **310'** in one-to-one correspondence. One end of each fan-out wiring **210'** is connected with the data signal line **110'**, and another end of each fan-out wiring **210'** is connected with an electrical signal of the data signal terminal **310'**.

(14) The data signal terminals **310'** are located in the driving area **300'** and each acts as a wiring end of a driving circuit **350'** in the driving area **300'**. The fan-out wirings **210'** are connected with an electrical signal of the driving circuit **350'** through the data signal terminals **310'**.

(15) The data signal lines **110'** are located in the display area **100'**, and the driving circuit **350'** is connected with the data signal lines **350'** through the fan-out wirings **210'**, and controls lighting of each sub-pixel in the display area **100'** through the data signal lines **110'**.

(16) The fan-out wirings **210'** are located in the fan-out area **200'**. In the prior art, one end of one fan-out wiring **210'** with straight ends and bent at a middle section is directly connected with the data signal line **110'**, and another end is directly connected with the data signal terminal **310'** to realize connection between the data signal terminal **310'** and the data signal line **110'**.

(17) Please refer to FIG. 2, the display apparatus includes a shell **10** and a display panel mounted on the shell **10**.

(18) Please refer to FIG. 3 in combination, the display panel **20** in the embodiment of the present application at least includes a display area **100**, a fan-out area **200**, and a driving area **300**. The fan-out area **200** is arranged on one side of the display area **100**, the driving area **300** is arranged on one side of the fan-out area **200** away from the display area **100**, and the fan-out area **200** is connected with the driving area **300** and the display area **100**. The display panel **20** further includes: a plurality of data signal lines **110**, the data signal lines **110** are located in the display area **100**; a plurality of data signal terminals **310**, the data signal terminals **310** are located in the driving area **300**; and a plurality of fan-out wirings **210**, the fan-out wirings **210** are located in the fan-out area **200**. The fan-out wirings **210** are connected with the data signal lines **110** and the data signal terminals **310** in one-to-one correspondence. Wherein, the fan-out wirings **210** at least include a first section wiring **211** connected with the data signal terminal **310**, a third section wiring **213** connected with the data signal line **110**, and a second section wiring **212** connected with the first section wiring **211** and the third section wiring **213**. The third section wiring **213** extends along a direction close to a central axis of the display panel **20** from the display area **100** to the driving area **300**. An extending direction of the first section wiring **211** and an extending direction of the second

section wiring **212** are different, and the extending direction of the second section wiring **212** and an extending direction of the third section wiring **213** are different.

(19) With the above-mentioned structure, the embodiment of the present application modifies the fan-out wiring **210'** with fewer bends to the fan-out wiring **210** with lots of bends, so that the fan-out wirings **210** are close to a central axis of the driving area, so that a difference between lengths of the first section wirings **211** on both sides and a length of the first section wiring **211** close to one side of the central axis is reduced, thus effectively shortening the length of the first section wiring **211** close to the one side of the central axis when the first section wiring **211** on both sides maintains a predetermined length, so that the fan-out area **200** is compressed, and the driving area **300** can be set closer to the display area **100**, which effectively shortens a width of the display panel border. To some extent, it solves a technical problem of the wiring mode in the prior art that affects narrowing of the lower border of a display, and realizes the narrowing of the lower border without affecting a wiring width and wiring resistance. After the fan-out area **200** is compressed, space on both sides of the fan-out wirings **210** increases, which can increase wiring space of other wirings (for example, CK clock signal wirings of chip on film (COF) coated film connected to gate on array (GOA) circuit), and appropriately increase the wiring width of other wirings to reduce the resistance.

(20) In some embodiments of the present application, an orthographic projection length **11** of each third section wiring **213** on a reference line L coplanar with the third section wiring **213** is less than a distance **12** between adjacent two data signal lines **110**. Wherein an extending direction of the reference line L is perpendicular to an extending direction of the central axis **351** of the display panel **20**, that is, the extending direction of the reference line is a horizontal transverse direction in FIG. 3.

(21) Such a wiring mode can prevent a density of the third section wiring **213** from being too large, so that the wiring width needs to be reduced, resulting in increased wiring resistance.

(22) In some embodiments of the present application, the display panel **20** further includes an electrostatic discharge protection area **400**. The electrostatic discharge protection area **400** is arranged on one side of the display area **100**, and the fan-out area **200** is arranged on one side of the electrostatic discharge protection area **400** away from the display area **100**.

(23) In some embodiments of the present application, the data signal line **110** includes a main portion **111** located in the display area **100** and a wiring portion **112** located in the electrostatic discharge protection area **400**. The wiring portion **112** is connected with the fan-out wiring **210**, and is connected with the driving circuit **350** through the fan-out wiring **210**, so that the driving circuit **350** controls the lighting of each sub-pixel in the display area **100** through the data signal line **110**. Wherein, the wiring portions **112** of at least two data signal lines **110** are of equal length.

(24) In some embodiments of the present application, please refer to FIG. 4 in combination, the above-mentioned fan-out wirings **210** further include a fourth section wiring **214**. The fourth section wiring **214** is connected with the second section wiring **212** and the third section wiring **213**. The fourth section wirings **214** of each fan-out wiring **210** are arranged in parallel along a first direction. The first direction is directly from the display area **100** to the driving area **300**, that is, a vertical direction.

(25) Specifically, one end of the fourth section wiring **214** is bent and connected with the second section wiring **212**, and another end is bent and connected with the third section wiring **213**, so that the second section wiring **212** is bent and connected with the third section wiring **213**.

(26) A first endpoint of the fourth section wiring **214** of each fan-out wiring **210** is located on a first straight line extending along a second direction, that is, a horizontal direction, and a second endpoint of the fourth section wiring **214** of each fan-out wiring **210** is located on a second straight line extending along the second direction, that is, the horizontal direction. Then the fourth section wiring **214** of each fan-out wiring **210** extends from the first straight line to the second straight line along the first direction, that is, the vertical direction; and then, the fourth section wirings **214** of

each fan-out wirings **210** are flush with each other and arranged in parallel with each other.

Wherein, the second direction intersects the first direction.

(27) The fourth section wiring **214** is used to calibrate endpoints of the second section wiring **212** and the third section wiring **213**, so as to ensure that the second section wiring **212** can be bent and connected with the third section wiring **213**.

(28) The fan-out wirings **210** are located in the fan-out area **200**, and the plurality of fan-out wirings **210** forms a plurality of fan-out wiring groups symmetrical about the central axis **351**, in the embodiment, including a first fan-out wiring group **201** and a second fan-out wiring group **202**. The first fan-out wiring group **201** and the second fan-out wiring group **202** are respectively distributed on both sides of the central axis **351** and symmetrical about the central axis **351**.

(29) In the first fan-out wiring group **201** and/or the second fan-out wiring group **202**, the second section wirings **212** of each fan-out wiring are arranged in parallel with each other. The central axis **351** coincides with a central line of the fan-out area **200** and the driving area **300**. Moreover, in the direction close to the central axis, lengths of the first section wirings **211** of at least two fan-out wirings increase progressively.

(30) In the first fan-out wiring group **201** and/or the second fan-out wiring group **202**, the first section wirings **211** of each fan-out wiring are arranged at equal intervals, the second section wirings **212** of each fan-out wiring are arranged at equal intervals, and the fourth section wirings **214** of each fan-out wiring are arranged at equal intervals.

(31) Through the above-mentioned fan-out wiring mode, the fan-out wirings can be more regular, which is conducive to further compressing the fan-out area.

(32) In some embodiments of the present application, a distance between the first section wiring **211** of the fan-out wiring **210** and the central axis **351**, a distance between the fourth section wiring **214** and the central axis **351**, and a distance between the wiring portion **112** and the central axis **351** decrease in sequence. The first section wiring **211** of the fan-out wiring **210** is closer to the central axis **351** than the corresponding connected wiring portion **112**.

(33) Specifically, a first distance between the first section wirings **211** of adjacent two fan-out wirings is less than a second distance between the fourth section wirings **214** of the adjacent two fan-out wirings. And the second distance is less than a third distance between the wiring portions **112** of adjacent two data signal lines **110**.

(34) In some embodiments of the present application, one end of the first section wiring **211** is connected with the data signal terminal **310**, and another end is bent and connected with the second section wiring **212**. A first fan-out angle α is formed through the first section wiring **211** bending and connecting the second section wiring **212**. One end of the wiring portion **112** is connected with the main portion **111** of the data signal line **110**, and another end is bent and connected with the third section wiring **213**. A second fan-out angle β is formed through the second section wiring **212** bending and connecting the third section wiring **213**. The second section wiring **212** is bent and connected with the third section wiring **213**. The first fan-out angle α and the second fan-out angle β are both obtuse angles, the first fan-out angle α is less than the second fan-out angle β . In this way, the fan-out area **200** can be further compressed to narrow the lower border.

(35) The first section wirings **211** of each fan-out wiring **210** are arranged along the first direction, that is, the vertical direction, consistent with an extending direction of the data signal line **110**. The length of each first section wiring **211** gradually increases along a direction from one side close to the central axis **351** to one side away from the central axis **351**.

(36) In some embodiments of the present application, a maximum length of the first section wiring **211** is 50 microns and a minimum length thereof is 10 microns. The first section wirings **211** of each fan-out wiring **210** belonging to a same fan-out wiring group are parallel to each other and arranged at equal intervals. The first distance between the first section wirings **211** of the adjacent two fan-out wirings **210** is 20 microns.

(37) The wiring portions **112** of each data signal line **110** are also arranged along the first direction,

that is, the vertical direction. The wiring portions **112** of each data signal line **110** are of a same length and aligned with each other. In some embodiments of the present application, the length of the wiring portion **112** is 20 microns. The wiring portions **112** of each fan-out wiring **210** belonging to the same fan-out wiring group are parallel to each other and arranged at equal intervals, and the third distance between the adjacent two wiring portions **112** is 90 microns.

(38) In some embodiments of the present application, the second section wirings **212** of each fan-out wiring **210** belonging to the same fan-out wiring group are parallel to each other and arranged at equal intervals, and the fourth distance between two adjacent second sections wirings **212** is 10 microns. The fourth distance is less than the first distance between the adjacent two first section wirings **211**, so that each fan-out wiring group takes up less space in the display area **100**, thus making the fan-out area **200** release more space.

(39) In some embodiments of the present application, the length of the fourth section wiring **214** is 5 microns. The fourth section wirings **214** of each fan-out wiring **210** belonging to the same fan-out wiring group are parallel to each other and arranged at equal intervals, and the second distance between the adjacent two fourth section wiring **214** is 25 microns. The second distance is greater than the first distance between the adjacent two first section wirings **211**, and less than the third distance between the adjacent two wiring portions **112**.

(40) In some embodiments of the present application, please refer to FIG. 5 in combination, the above-mentioned display panel further includes a plurality of common signal lines **120**, a plurality of common signal terminals **320**, and a plurality of common wirings **220**. The common signal lines **120** and the common signal terminals **320** are in one-to-one correspondence with the common wirings **220**. One end of each common wiring **220** is connected with the common signal line **120**, and another end is connected with an electrical signal of the common signal terminal **320**.

(41) The common signal terminals **320** are located in the driving area **300**, and each of them acts as a wiring end of the driving circuit **350** in the driving area **300**. The common wirings **220** are connected with an electrical signal of the driving circuit **350** through the common signal terminals **320**. The common signal terminals **320** are vertically arranged on one side of the driving circuit **350** in parallel, which are also located on one side of the driving circuit **350** facing the display area **100**, and on both sides of the data signal terminals **310**. In other embodiments, the common signal terminals **320** can also be arranged on one side of the driving circuit **350** away from the display area **100**.

(42) The common signal lines **120** are located in the display area **100**, and the common signal lines **120** are connected with an electrical signal of a common electrode in the display area **100**.

(43) The common wirings **220** are located in the fan-out area **200**. One end of each common wiring **220** is connected with the common signal terminal **320**, another end is connected with the common signal line **120**, so as to realize electrical connection between the common signal terminal **320** and the common signal line **120**.

(44) The plurality of common wirings **220** form a plurality of common wiring groups **203**. The common wiring groups **203** are arranged on side of the first fan-out wiring group **201** away from the second fan-out wiring group **202** and one side of the second fan-out wiring group **202** away from the first fan-out wiring group **201**. The driving circuit **350** can transmit a common signal to the common electrode in the display area **100** through the common wiring groups **203**.

(45) In other embodiments of the present application, please refer to FIG. 6 in combination, the above-mentioned display panel further includes a signal conversion circuit **240**, a plurality of conversion wirings **250**, a plurality of control signal lines **130**, a plurality of control signal terminals **330**, and a plurality of signal control wirings **230**. The common signal lines **120** and the common signal terminals **320** are in one-to-one correspondence with the common wirings **220**. One end of each common wiring **220** is connected with the common signal line **120**, and another end is connected with the electrical signal of the common signal terminal **320**.

(46) The control signal terminals **330** are located in the driving area **300**, and each of them acts as a

wiring end of the driving circuit **350** in the driving area **300**. The common wirings **220** are connected with the electrical signal of the driving circuit **350** through the control signal terminals **330**. The control signal terminals **330** are vertically arranged on one side of the driving circuit **350** in parallel, which are also located on one side of the driving circuit **350** facing the display area **100**, and on both sides of the data signal terminals **310**. In other embodiments, the control signal terminals **330** can also be arranged on one side of the driving circuit **350** away from the display area **100**.

(47) The control signal lines **130** are located in the display area **100**, and are connected with a corresponding control circuit in the display area **100**.

(48) The signal conversion circuit **240**, the conversion wirings **250**, and the signal control wirings **230** are located in the fan-out area **200**.

(49) The plurality of signal control wirings **230** form a plurality of signal control wiring groups **204**. The signal control wiring groups **204** and the signal conversion circuit **240** are both arranged on one side of the first fan-out wiring group **201** away from the second fan-out wiring group **202** and one side of the second fan-out wiring group **202** away from the first fan-out wiring group **201**. The driving circuit **350** can transmit a control signal to the control circuit in the display area **100** through the signal control wiring groups **204**.

(50) The signal conversion circuit **240** is electrically connected with the control signal terminals **330** of the driving circuit **350** through the conversion wirings **250** to convert the electrical signal of the driving circuit **350** into a control signal that can be executed in the display area **100**. One end of each signal control wiring **230** is connected with the signal conversion circuit **240**, and another end is connected with the control signal line **130**, so as to realize electrical connection between the control signal terminal **330** and the control signal line **130**. The driving circuit **350** can transmit a signal-control-signal to the display area **100** through the signal control wiring groups **204**. The signal-control-signal includes a touch signal, a data distribution signal, or a clock signal, etc.

(51) In above-mentioned embodiments, a bent arrangement of the fan-out wirings **210** makes the fan-out wirings **210** close to the central axis **351**, leaving a large amount of space on both sides of the fan-out area **200**, which can be used to set various lines and components, so that the fan-out area **200** can accommodate more lines and components in the limited space, increasing functions that can be realized by the display panel, and improving space utilization of the fan-out area **200**.

(52) In some embodiments, in the driving area **300**, a plurality of driving circuits **350** can be set, and a plurality of driving circuits **350** are arranged at intervals along one side of the display area **100**. Accordingly, the display area **100** is also divided into a plurality of sub-display areas along one side opposite to the driving circuits **350**. Each sub-display area corresponds to one driving circuit **350** and is connected with the driving circuit **350** through the corresponding fan-out area **200**.

(53) For each driving circuit **350** and its corresponding sub-display area, a corresponding fan-out wiring group and other wiring groups are provided to connect the driving circuit **350** and the corresponding sub display area. Other wirings include the above-mentioned common wiring groups **203** or signal control wiring groups **204**. In the fan-out area **200** corresponding to each sub-display area, the fan-out wiring groups are symmetrically arranged along the central axis **351**, and other wiring groups are arranged on both sides of the fan-out wiring groups.

(54) A display apparatus is provided in the embodiments of the present application, and the display apparatus includes a shell and the display panel provided in the above-mentioned embodiments. Wherein, the display panel is mounted on the shell.

(55) In the display apparatus of the embodiments of the present application, the fan-out wiring **210'** with fewer bends are modified to the fan-out wiring **210** with lots of bends, so that the fan-out wirings **210** are close to the central axis of the driving area, so that a difference between lengths of the first section wirings **211** on both sides and a length of the first section wiring **211** close to one side of the central axis is reduced, thus effectively shortening the length of the first section wiring

211 close to the one side of the central axis when the first section wiring 211 on both sides maintains the predetermined length, so that the fan-out area 200 is compressed, and the driving area 300 can be set closer to the display area 100, which effectively shortens a width of the display panel border. To some extent, it solves a technical problem of the wiring mode in the prior art that affects narrowing of the lower border of the display, and realizes the narrowing of the lower border without affecting a wiring width and wiring resistance. After the fan-out area 200 is compressed, space on both sides of the fan-out wirings 210 increases, which can increase wiring space of other wirings, and appropriately increase the wiring width of other wirings to reduce the resistance.

(56) The display apparatus and the display panel thereof provided by the embodiments of the present application are described in detail above. Specific examples are used in this paper to illustrate principles and implementations of this disclosure, description of the above embodiments are only used to help understand method of the present application and its core idea; meanwhile, for those skilled in the art, according to idea of the present application, there will be changes in specific embodiments and scope of application, in conclusion, contents of this specification should not be construed as limiting the present application.

Claims

1. A display panel, comprising a display area, a fan-out area, and a driving area; the fan-out area being arranged on one side of the display area, the driving area being arranged on one side of the fan-out area away from the display area, and the fan-out area being connected with the driving area and the display area; the display panel further comprising: a plurality of data signal lines located in the display area; a plurality of data signal terminals located in the driving area; and a plurality of fan-out wirings located in the fan-out area; the fan-out wirings being connected with the data signal lines and the data signal terminals in one-to-one correspondence; wherein, the fan-out wirings at least comprise a first section wiring, a second section wiring, and a third section wiring; the first section wiring is connected with the data signal terminals, the third section wiring is connected with the data signal lines, and the second section wiring is connected with the first section wiring and the third section wiring; the third section wiring extends from the display area to the driving area along a direction at an oblique angle to a horizontal line perpendicular to a central axis of the display panel, an extending direction of the first section wiring and an extending direction of the second section wiring are different, and the extending direction of the second section wiring and an extending direction of the third section wiring are different; wherein, each fan-out wiring further comprises a fourth section wiring, and the fourth section wiring is connected with the second section wiring and the third section wiring; the fourth section wiring of each fan-out wiring is arranged in parallel along a first direction, and the first direction is directed from the display area to the driving area.
2. The display panel as claimed in claim 1, wherein, an orthogonal projection length of each third section wiring on a reference line coplanar with the third section wiring is less than a distance between adjacent two data signal lines; wherein an extending direction of the reference line is perpendicular to an extending direction of the central axis of the display panel.
3. The display panel as claimed in claim 1, wherein, a first endpoint of the fourth section wiring of each fan-out wiring is located on a first straight line extending along a second direction, a second endpoint of the fourth section wiring of each fan-out wiring is located on a second straight line extending along the second direction, and the second direction intersects the first direction.
4. The display panel as claimed in claim 3, wherein, the plurality of fan-out wirings comprise a plurality of fan-out wiring groups symmetrical about the central axis of the display panel; in at least one fan-out wiring group, the second section wiring of each fan-out wiring is arranged in parallel, and the central axis coincides with a central line of the fan-out area and the driving area.
5. The display panel as claimed in claim 4, wherein, in at least one fan-out wiring group, in the

direction pointing from the side of the display panel away from the central axis of the display panel to the central axis, lengths of the first section wirings of at least two fan-out wirings increase progressively.

6. The display panel as claimed in claim 4, wherein, in at least one fan-out wiring group, the first section wirings of the fan-out wirings are arranged at equal intervals, the second section wirings of the fan-out wirings are arranged at equal intervals, and the fourth section wirings of the fan-out wirings are arranged at equal intervals.

7. The display panel as claimed in claim 6, wherein, in at least one fan-out wiring group, a first distance between the first section wirings of adjacent two fan-out wirings is less than a second distance between the fourth section wirings of the adjacent two fan-out wirings, and the second distance is less than a third distance between wiring portions of the adjacent two fan-out wirings.

8. The display panel as claimed in claim 1, wherein, a fourth distance of the second section wirings of adjacent two fan-out wirings is less than a first distance of the first section wirings of the adjacent two fan-out wirings.

9. A display apparatus, wherein, comprising: a shell; a display panel mounted on the shell and at least comprising a display area, a fan-out area, and a driving area; the fan-out area being arranged on one side of the display area, the driving area being arranged on one side of the fan-out area away from the display area, and the fan-out area being connected with the driving area and the display area; the display panel further comprising: a plurality of data signal lines located in the display area; a plurality of data signal terminals located in the driving area; and a plurality of fan-out wirings located in the fan-out area; the fan-out wirings being connected with the data signal lines and the data signal terminals in one-to-one correspondence; wherein, the fan-out wirings at least comprise a first section wiring, a second section wiring, and a third section wiring; the first section wiring is connected with the data signal terminals, the third section wiring is connected with the data signal lines, and the second section wiring is connected with the first section wiring and the third section wiring; the third section wiring extends from the display area to the driving area along a direction at an oblique angle to a horizontal line perpendicular to a central axis of the display panel, an extending direction of the first section wiring and an extending direction of the second section wiring are different, and the extending direction of the second section wiring and an extending direction of the third section wiring are different; wherein, each fan-out wiring further comprises a fourth section wiring, and the fourth section wiring is connected with the second section wiring and the third section wiring; the fourth section wiring of each fan-out wiring is arranged in parallel along a first direction, and the first direction is directed from the display area to the driving area.

10. The display apparatus as claimed in claim 9, wherein, an orthogonal projection length of each third section wiring on a reference line coplanar with the third section wiring is less than a distance between adjacent two data signal lines; wherein an extending direction of the reference line is perpendicular to an extending direction of the central axis of the display panel.

11. The display apparatus as claimed in claim 9, wherein, a first endpoint of the fourth section wiring of each fan-out wiring is located on a first straight line extending along a second direction, a second endpoint of the fourth section wiring of each fan-out wiring is located on a second straight line extending along the second direction, and the second direction intersects the first direction.

12. The display apparatus as claimed in claim 11, wherein, the plurality of fan-out wirings comprise a plurality of fan-out wiring groups symmetrical about the central axis of the display panel; in at least one fan-out wiring group, the second section wiring of each fan-out wiring is arranged in parallel, and the central axis coincides with a central line of the fan-out area and the driving area.

13. The display apparatus as claimed in claim 12, wherein, in at least one fan-out wiring group, in the direction pointing from the side of the display panel away from the central axis of the display panel to the central axis, lengths of the first section wirings of at least two fan-out wirings increase progressively.

14. The display apparatus as claimed in claim 12, wherein, in at least one fan-out wiring group, the first section wirings of the fan-out wirings are arranged at equal intervals, the second section wirings of the fan-out wirings are arranged at equal intervals, and the fourth section wirings of the fan-out wirings are arranged at equal intervals.

15. The display apparatus as claimed in claim 14, wherein, in at least one fan-out wiring group, a first distance between the first section wirings of adjacent two fan-out wirings is less than a second distance between the fourth section wirings of the adjacent two fan-out wirings, and the second distance is less than a third distance between wiring portions of the adjacent two fan-out wirings.

16. The display apparatus as claimed in claim 9, wherein, a fourth distance of the second section wirings of adjacent two fan-out wirings is less than a first distance of the first section wirings of the adjacent two fan-out wirings.
